Sec: JEE APEX & SR IIT MODEL-A | DT: 12 /12 /2025
Time: 3 Hrs | GTM – 05 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
C	**B**	**B**	**D**	**D**	**B**	**A**	**B**	**D**	**B**
11	12	13	14	15	16	17	18	19	20
A	**B**	**A**	**A**	**A**	**B**	**C**	**B**	**A**	**C**
21	22	23	24	25					
64	**6**	**5**	**30**	**2**					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	**A**	**A**	**C**	**B**	**A**	**C**	**A**	**B**	**C**
36	37	38	39	40	41	42	43	44	45
D	**A**	**B**	**A**	**C**	**D**	**C**	**A**	**B**	**D**
46	47	48	49	50					
4	**2**	**45**	**6**	**68**					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	**C**	**A**	**D**	**A**	**C**	**C**	**C**	**D**	**C**
61	62	63	64	65	66	67	68	69	70
D	**B**	**C**	**C**	**B**	**C**	**D**	**A**	**D**	**B**
71	72	73	74	75					
8	**51**	**3**	**15**	**6**					

Rankers Academy JEE
JEE-MAIN 2026
For More Join: @IITJEE_Advanc

HINTS
PHYSICS

1. $E = 7.5\,V/cm = 750\,V/m$

Force : $F = eE = 1.6\times10^{-19}.750 = 1.2\times10^{-16}\,N$

Acceleration: $a = \dfrac{F}{m} = \dfrac{1.2\times10^{-16}}{9.1\times10^{-31}} = 1.32\times10^{14}$

Time inside plates: $t = \dfrac{0.15}{2\times10^{6}} = 7.5\times10^{-8}\,s$

Vertical velocity: $v_y = at = 1.32\times10^{14}.\left(7.5\times10^{-8}\right) = 9.9\times10^{6}\ m/s$

2. Conceptual

3. Mass removed $m = \dfrac{\pi\left(R/3\right)^2}{\pi R^2}M = \dfrac{1}{9}M$

MOI of removed disc about its own centre:

$$I_{cm} = \frac{1}{2}m\left(R/3\right)^2 = \frac{1}{2}.\frac{M}{9}.\frac{R^2}{9} = \frac{MR^2}{162}$$

Distance from centre of the bigger disc is $d = 2R/3$.

Using parallel axis theorem:

$$\frac{MR^2}{162} + \frac{M}{9}.\frac{4R^2}{9} = \frac{MR^2}{162} + \frac{4MR^2}{81} = \frac{9MR^2}{162} = \frac{1}{18}MR^2$$

Big disc MOI: $I_{big} = \dfrac{1}{2}MR^2$

Remaining MOI: $\dfrac{1}{2}MR^2 - \dfrac{1}{18}MR^2 = \dfrac{9-1}{18}MR^2 = \dfrac{8}{18}MR^2 = \dfrac{4}{9}MR^2$

4. Heat absorbed by ice to $0°C$

$Q_1 = 0.002\times2100\times5 = 21J$

For melting of ice:

$Q_2 = 0.002\times3.3\times10^{5} = 660\ J$

Heat for water $0^0C \rightarrow 100°C$

For More Join: @IITJEE_Advanc

$Q_3 = 0.002 \times 4200 \times 100 = 840\,J$

For Vaporisation:

$Q_4 = 0.002 \times 2.25 \times 10^6 = 4500\ J$

Heat for steam $100^0 C \rightarrow 120°C$

$Q_5 = 0.002 \times 2000 \times 20 = 80J$

Total : $Q = 21 + 660 + 840 + 4500 + 80 = 6101J$

5. $\frac{r_n}{r_1} = \frac{4.68 \times 10^{-10}}{5.2 \times 10^{-11}} = 9$

Thus $n^2 = 9 \Rightarrow n = 3.$

Bohr model: $\lambda_n \propto n$

So ratio $= 3$

6. Total energy at bottom $E = 3mgl$

Height at angle θ:

$h = l(1 - \cos\theta)$

For $B, \theta = 45°$:

$h_B = l\left(1 - \frac{1}{\sqrt{2}}\right)$

For $C, \theta = 135°$:

$h_C = l\left(1 + \frac{1}{\sqrt{2}}\right)$

Kinetic energies:

$K_B = E - mgh_B$

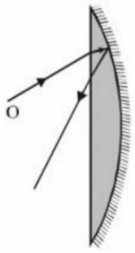
$K_C = E - mgh_C$

Compute ratio (after simplification)

$K_B : K_C = 2:1$

7. $P = 2P_L + P_M$

For More Join: @IITJEE_Advanc



$$\Rightarrow \frac{1}{F}=\frac{2}{f_l}+\frac{1}{f_m} \Rightarrow \frac{1}{F}=\frac{2(\mu-1)}{R}+\frac{2}{R}$$

$$\Rightarrow \frac{1}{F}=\frac{2\mu}{R} \Rightarrow F=\frac{R}{2\mu}$$

8. Conceptual

9. $\dfrac{\dfrac{5}{30(x)}}{30+x}=\dfrac{10}{20}$

$x=15\Omega$

$2(10+20)=i(5+10)$

$i=4A$

$i_x=\dfrac{i(30)}{30+x}=\dfrac{4(30)}{30+15}=\dfrac{8}{3}A$

10. Gravitation field at 'm' due to hollowed – out lead sphere

$=\{\text{Field due to solid spere}\}-\{\text{Field due to mass that was removed}\}$

Field due to solid sphere $=\dfrac{GM}{d^2}=E_1=\dfrac{GM}{4R^2}$

Field due to removed mass $=\dfrac{GM'}{x^2}=E_2$

$M'=\dfrac{M}{\frac{4}{3}\pi R^3}\times\dfrac{4}{3}\pi\left(\dfrac{R}{2}\right)^3=\dfrac{M}{8}$ and $x=d-\dfrac{R}{2}$

For More Join: @IITJEE_Advanc

So, $E_2 = \dfrac{GM}{8\left(d-\dfrac{R}{2}\right)^2} = \dfrac{GM}{8\left(\dfrac{3R}{2}\right)^2} = \dfrac{GM}{18R^2}$

$$E_{net} = E_1 - E_2 = \frac{GM}{R^2}\left\{\frac{1}{4} - \frac{1}{18}\right\} = \frac{7GM}{36R^2}$$

$$F_{net} = mE_{net} = \frac{7GMm}{36R^2}$$

11. Both pipes vibrate in first overtone.

$$f_C = f_O$$

$$3\left(V_C / 4L\right) = 2\left(V_o / \left(2l_o\right)\right)$$

$$\therefore \left(3/2\right)\left(V_C / L\right) = \left(V_O / l_O\right)$$

$$\therefore \left(V_O / V_C\right) = \left(3/2\right)\left(l_O / L\right)$$

also $\left(V_O / V_C\right) = \sqrt{\left(\dfrac{\rho_1}{\rho_2}\right)}$ (elastic moduli are same)

$$\therefore \sqrt{\left(\rho_1 / \rho_2\right)} = \left(3/4\right)\left(l_0 / L\right)$$

$$\therefore l_O = \left(4L/3\right)\sqrt{\left(\rho_1 / \sqrt{\rho_2}\right)}$$

12. $E = \dfrac{KQ}{R^2}$

$$E = \frac{Q}{4\pi\varepsilon_0 R^2}$$

$$\varepsilon_0 = \frac{Q}{4\pi R^2 E}$$

Now, $\varepsilon_0 E^2 = \dfrac{Q}{4\pi R^2 E}.E^2 = \dfrac{Q}{4\pi R^2}.E$

$$\left[\varepsilon_0 E^2\right] = \left[\frac{QE}{R^2}\right] = \left[\frac{F}{R^2}\right]$$

For More Join: @IITJEE_Advanc

13. Total enclosed charge, $q = 100Q$ coulomb

$$\phi_E = \frac{q}{\varepsilon_0} = \frac{100Q}{\varepsilon_0}$$

14. Conceptual

15. Photon energy:

$$E = \frac{1240}{480} \approx 2.58 eV$$

Compare with work functions:

$Na: 2.3\ eV \rightarrow 2.58 > 2.3 \rightarrow$ Yes

$K: 2.0\ eV \rightarrow 2.58 > 2.0 \rightarrow$ Yes

Photoemission from both metals

16. Radiated power: $P \propto R^2 T^4$

Ratio: $\frac{P_L}{P_S} = \frac{(0.9)^2 (350)^4}{(0.3)^2 (700)^4}$

17. Net power = $= P_1 + P_2 + P_3$

$$= \frac{(\mu_1 - 1)}{R_1} + (\mu_2 - 1)\left(\frac{1}{-R_1} + \frac{1}{-R_2}\right) + \frac{(\mu_3 - 1)}{R_2}$$

$$= \frac{(\mu_1 - \mu_2)}{R_1} + \frac{(\mu_3 - \mu_2)}{R_2}$$

$$= \left(\frac{4}{3} - \frac{3}{2}\right)\frac{1}{R_1} + \left(\frac{4}{3} - \frac{3}{2}\right)\frac{1}{R_2}$$

$$= -\frac{1}{6}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = -\left(\frac{R_1 + R_2}{6R_1R_2}\right)$$

18. $\beta = \frac{\lambda D}{d}$

In water, λ decreases $\rightarrow \beta$ decreases.

Both statements are true, and R correctly explains A.

For More Join: @IITJEE_Advanc

19. Initial charge:

$$Q_i = CV = 30\mu F \times 200 = 6mC$$

New capacitance:

$$C_f = KC = 90\mu F$$

New charge:

$$Q_f = 90\mu f \times 200 = 18mC$$

Extra charge:

$$\Delta Q = 12mC$$

Energy change:

$$\Delta U = \frac{1}{2}C_f V^2 - \frac{1}{2}C_i V^2 = \frac{1}{2}(90-30)\mu F \times (200)^2$$

$$= \frac{1}{2} \times 60 \times 10^{-6} \times 40000 = 1.2\ J$$

20. Conceptual

21. For a spherical mirror: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow v = \frac{uf}{u-f}$

Differentiate w.r.t. time (f constant) and use $|du/dt| = 20$

$$a = \frac{d^2v}{dt^2} = \frac{2f^2}{(u-f)^3}\left(\frac{du}{dt}\right)^2, \text{ with}$$

$$f = 1, u = -4, (u-f) = -5:$$

$$a = \frac{2.1^2.(20)^2}{(-5)^3} = \frac{800}{-125} = -6.4\ m/s^2.$$

22. For bubbles in contact (curvature toward smaller bubble):

$$\frac{1}{R} = \frac{1}{r_1} - \frac{1}{r_2} = \frac{1}{3} - \frac{1}{6} = \frac{1}{6} \Rightarrow R = 6\ cm$$

23. Compute at $t = 1$:

$$\vec{r}_A = (2,-1,0),\ \vec{r}_B = (1,2,0)$$

For More Join: @IITJEE_Advanc

Relative position $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (1,-3,0)$.

Velocity of $A: \vec{v}_A = (2,-1,0)$. (Mass=1 kg.)

Angular momentum: $\vec{L} = \vec{r}_{AB} \times \vec{v}_A$.

$L = 1.(-1) - (-3).2 = -1 + 6 = 5$.

Magnitude $|\vec{L}| = 5\ kg.m^2/s$.

24.

l	l	l
A K	A $2K$	A $3K$

$\frac{Q}{t} = same$

$\left(\frac{Q}{t}\right)_{\text{combinations}} = \frac{K^1 A\Delta\theta}{3l}$

$\frac{3l}{K^1} = \frac{l}{K} + \frac{l}{2K} + \frac{l}{3K}$

$\frac{3}{K^1} = \frac{6+3+2}{K} = \frac{11}{6K}$

$K = \frac{18}{11}K$

$\left(\frac{Q}{t}\right)_{comb} = \frac{18KA(110)}{11(3l)}$

$= \frac{60KA}{l}$

$\left(\frac{Q}{t}\right)_{middle} = \frac{2KA(\Delta\theta)}{l}$

$\left(\frac{Q}{t}\right)_{com} = \left(\frac{Q}{t}\right)_{middle}$

For More Join: @IITJEE_Advanc

$$\frac{60KA}{l}=\frac{2KA}{l}\Delta\theta \qquad \Delta\theta=30^0C$$

25. Vertical component

$$v_y=u\sin 30^\circ=30\times\frac{1}{2}=15\ m/s$$

Height in first second:

$$h_0=u_y(1)-\frac{1}{2}g(1)^2=15-5=10m$$

Height gained in last second before top before reaching the highest point is always $h_1=\frac{1}{2}g=5m$ (standard result)

So ratio $h_0:h_1=10:5=2:1$

CHEMISTRY

26. R = 0.529 x $\frac{n^2}{Z}$

$$(r_{Li}2^+)_{n=2}=0.529\times\frac{(2)^2}{3}=x$$

$$(r_{Be}3^+)_{n=3}=0.529\times\frac{(3)^2}{4}$$

$$\frac{(r_{Li}2^+)_{n=2}}{(r_{Be}3^+)_{n=3}}=\frac{\frac{r_o\times(2)^2}{3}}{\frac{r_o\times(3)^2}{4}}$$

$$\frac{x}{(r_{Be}3^+)_{n=3}}=\frac{16}{27}$$

$$(r_{Be}3^+)_{n=3}=\frac{27x}{16}$$

27. Cis-1,2-dichloroethene has higher dipole moment than trans-1,2-dichloroethene

28. Conceptual

29. Conceptual

For More Join: @IITJEE_Advanc

30. Only Lu^{3+} & Ce^{4+} are diamagnetic due to absence of unpaired electrons.
Remaining all ions have partially filled f-orbitals.

31. Conceptual

32. Conceptual

33. KBr on electrolysis gives H_2 at cathode and Br_2 at anode.
CH_3COONa on electrolysis gives H_2 & NaOH at cathode and C_2H_6 & CO_2 at anode.
$AgNO_3$ on electrolysis gives Ag at cathode and O_2 at anode.

34. Conceptual

35. Tryptophan is an amino acid

36. Atomic radius; Tl > In > Al > Ga > B
Denisity ; B < Al < Ga < In < Tl
Melting point ; B > Al > Tl > In > Ga
Boiling point ; B > Al > Ga > In > Tl

37. $K = \frac{2.303}{t}.log\frac{a}{a-x}$

$t_{1/2} = \frac{0.693}{K}$

38. Conceptual

39. Conceptual

40. According to lechateliers principle

41. Volume of O_2 evolved at STP, $V_{O_2} = \frac{E_{V_{O_2}} \times i \times t}{F} = \frac{5600 \times i \times 1}{96500}$

$i = \frac{0.224 \times 96500}{5600}$ =3.38 Amps

42. Conceptual

43. The value of Δ_0 depends on strength of ligand, oxidation state and position of metal ion in periodic table.

44. The correct order of boiling points is in increasing order of im value.
i. e., 3 < 4 < 1 < 2

For More Join: @IITJEE_Advanc

45. A) $\left[Ni\left(H_2O\right)_6\right]^{2+}$ sp^3d^2, paramagnetic

B) $\left[PtC\ell_6\right]^{2-}$ d^2sp^3, diamagnetic

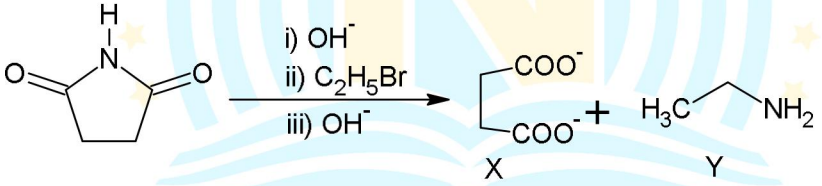
C) $\left[Ni\left(CO\right)_4\right]$; sp^3, diamagnetic

46. $x = 2(1,4)$ $y = 2(7,9)$

$x + y = 2 + 2 = 4$

47. $\log\frac{K_2}{K_1} = \frac{E_0}{2.303R}\left(\frac{T_2 - T_1}{T_2T_1}\right)$. Find E_a for 7°C and 17°C pair & Use it for 17°C to 27°C pair

48.



49. $CO_2 + 2KOH \rightarrow K_2CO_3 + H_2O$

No. of moles of CO_2 formed = $\frac{300 \times 2 \times 10^{-3}}{2} = \frac{0.6}{2} = 0.3$

$C_nH_{zn+2}\left(\frac{3n+1}{2}\right)O_2 \longrightarrow n_{CO_2} + (n+1)H_2O$

1 mole → n moles

$\frac{4.3}{(12n+2n+2)}$ mole → ?

$\frac{4.3}{14n+2}.n = 0.3$

$4.3n = 4.2n + 0.6$

$0.1n = 0.6$

$n = 6$

For More Join: @IITJEE_Advanc

50. $\%\ \text{of Br} = \dfrac{\text{Atomic mass of Brxm,}}{\text{molecular mass of AgBr} \times \text{m}} \times 100$

M_1 = mass of organic compound taken

M_1 = mass of AgBr obtained

$$\therefore \%\ of\ Br = \frac{80 \times 0.2397}{188 \times 0.15} \times 100$$
$$= 68\%$$

MATHEMATICS

51. $r = 2, \sum_{n=1}^{100} a_n = 150$

52. $\dfrac{3dy}{dx} + \dfrac{2y}{x+1} = \dfrac{x^3}{y^2}$

$$\Rightarrow \frac{3y^2 dy}{dx} + \frac{2y^3}{x+1} = x^3$$

Let $y^3 = t$

$$\Rightarrow \frac{3y^2 dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dt}{dx} + \frac{2t}{x+1} = x^3$$

I.F= $e^{\int \frac{2}{x+1} dx} dx = e^{2\ln(1+x)} = (1+x)^2$

$$\Rightarrow t(x+1)^2 = \int x^3 (x+1)^2 dx = \int x^5 + x^3 + 2x^4 dx = \frac{x^6}{6} + \frac{x^4}{4} + \frac{2x^5}{5} + k$$

53. M: event that all the materials will be delivered at the correct time.

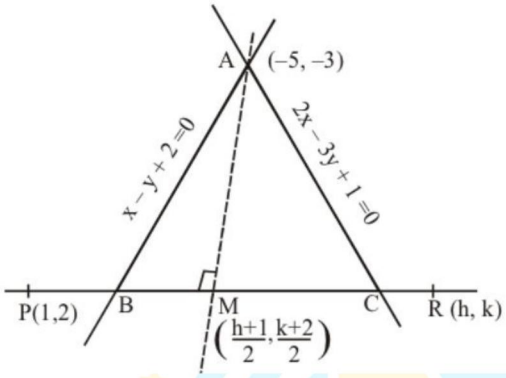
F: event that the building programme will be completed on time.

$$P\left(\frac{F}{M'}\right) = \frac{P(F \cap M')}{P(M')} = \frac{P(F) - P(F \cap M)}{1 - P(M)} = \frac{0.7 - 0.65}{1 - 0.8} = \frac{1}{4}$$

54. Put, $log_3 x = t$

For More Join: @IITJEE_Advanc

55.



$$\frac{k-2}{h-1}\times\frac{\left(\frac{k+2}{2}+3\right)}{\left(\frac{h+1}{2}+5\right)}=-1$$

$$\Rightarrow h^2+k^2+10h+6k-27=0$$

Therefore the required locus is

$$x^2+y^2+10x+6y-27=0$$

56. Let $I=\int_0^1\left(1-x^{50}\right)^{100}dx$ and $I'=\int_0^1\left(1-x^{50}\right)^{101}dx$

Then, $I'=\int_0^1 1.\left(1-x^{50}\right)^{101}dx$

$$=\left[x\left(1-x^{50}\right)^{101}\right]_0^1+101\int_0^1 50x^{50}\left(1-x^{50}\right)^{100}d$$

$$=+5050\int_0^1 x^{50}\left(1-x^{50}\right)^{100}dx$$

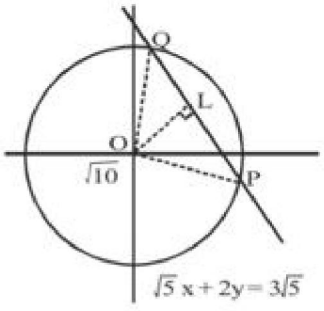
$$-I'=+5050\int_0^1 -x^{50}\left(1-x^{50}\right)^{100}dx=5050\int_0^1\left(-1+1-x^{50}\right)\left(1-x^{50}\right)^{100}$$

$$-I'=-5050I+5050I'\Rightarrow\frac{I}{I'}=\frac{5051}{5050}$$

For More Join: @IITJEE_Advanc

57. Length of perpendicular from origin to the line $x\sqrt{5}+2y=3\sqrt{5}$ is

$$OL=\frac{3\sqrt{5}}{\sqrt{\left(\sqrt{5}\right)^2+2^2}}=\frac{3\sqrt{5}}{\sqrt{9}}=\sqrt{5}$$



Radius of the given circle $=\sqrt{10}=OQ=OP$

$$PQ=2QL=2\sqrt{OQ^2-OL^2}=2\sqrt{10-5}=2\sqrt{5}$$

Thus area of $\Delta OPQ=\frac{1}{2}\times PQ\times OL=\frac{1}{2}\times 2\sqrt{5}\times\sqrt{5}=5$

58. Shortest distance between the lines (skew lines)

59. $f\left(x+2y\right)=f\left(x\right)+f\left(2y\right)+4xy\ \ \forall\ x,y\in R$

Replace $2y$ with y we have

$f\left(x+y\right)=f\left(x\right)+f\left(y\right)+2xy\ \forall x,y\in R$

Diff. w.r.t. $x: f'\left(x+y\right)=f'\left(x\right)+2y$

Put $x=1,\ y=-1\quad\Rightarrow f'\left(0\right)=f'\left(1\right)-2$

60. First we select n grand children from $2n$ grand children is ${}^{2n}C_n$

Now arrangement of both group is $n!\times n!$

Now rest all $\left(m+1\right)$ place where we occupy the grandfather and m sons but grandfather refuse the sit to either side of grand children so the out of $m-1$ seat one seat can be selected

Now required number of sitting in ${}^{2n}C_n\times n!\times n!\times{}^{(m-1)}C_1.m!$

For More Join: @IITJEE_Advanc

$$=\frac{2n!}{n!.n!}.n!.n!.{}^{(m-1)}C_1.m!=2n!.m!.(m-1)$$

61. Minimum values is $\frac{3\pi}{2}$, maximum value is 3π

Product is $\frac{3\pi}{2}.3\pi=\frac{9\pi^2}{2}$

62. $f(x+1)=f(x)+2x+1;$

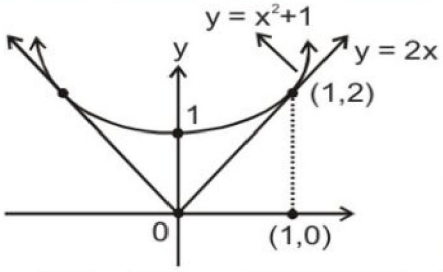
$f(x)\rightarrow$ Polynomial function

With $f(0)=1$

Degree of $f(x)=2$

Also,

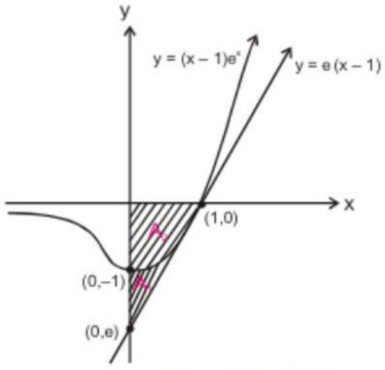
$$\left.\begin{array}{r} f(0)=1 \\ x=0: f(1)=2 \\ x=1: f(2)=5 \\ x=2: f(3)=10 \end{array}\right] f(x)=x^2+1$$



$$A=2\left[\int_0^1\left(x^2+1\right)dx-1\right]=\frac{2}{3}$$

For More Join: @IITJEE_Advanc

63.



$$A_1 = \frac{3}{2} - A_2 \Rightarrow \qquad A_1 + A_2 = \frac{e}{2}$$

$$A_2 = -\int_0^1 (x-1)e^x dx = -\left[(x-1)e^x - e^x\right]_0^1$$

$$\therefore A_2 = -\left[-3-(-1-1)\right] = e-2$$

$$\therefore A_1 = \frac{e}{2} - (e-2) = 2 - \frac{e}{2}$$

64. Distance between vertex and focus is $a(e-1) = 4 \Rightarrow a = 4\ (e = 2)$

$$\therefore b^2 = a^2(e^2-1) = 48$$

Distance of vertex from centre is a

$\therefore$ Coordinates of the centre is $(-2,3)$

$\therefore$ Equation of the required hyperbola is $\dfrac{(x+2)^2}{16} - \dfrac{(y-3)^2}{48} = 1$

65. $t_n = S_n - S_{n-1} = \dfrac{n}{6}(2n^2+9n+13) - \dfrac{(n-1)}{6}\left\{2(n-1)^2+9(n-1)+13\right\}$

$$= \frac{1}{6}\left[2n^3+9n^2+13n-2(n-1)^3-9(n-1)^2-13(n-1)\right] = (n+1)^2$$

$$\sum_{r=1}^{100} r.(r+1) = \frac{100\times101\times102}{3} = 343400$$

For More Join: @IITJEE_Advanc

66. Conceptual

67. a_1 divides b_2

Each element has 2 choices

$\Rightarrow 3\times 2=6$

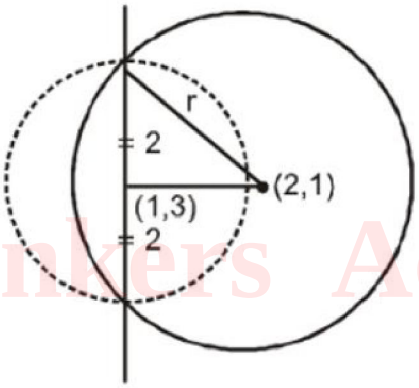
a_2 divides b_1

Each element has 2 choices

$\Rightarrow 3\times 2=6$

Total $=6\times 6=36$

68. $r^2=2^2+\left\{(2-1)^2+(1-3)^2\right\}\Rightarrow r=3$



69. The numbers upto 25 which gives g.c.d with 24 equals to 1 are

$1,5,7,11,13,17,19$ and 23.

Sum of these numbers $=96$

There are four such blocks and a number 97 is there upto 100.

$\therefore$ Complete sum

$=96+(24\times 8+96)+(48\times 8+96)+(72\times 8+96)+97$

$=1633$

70. Let $z=r_1e^{i\theta}$ and $w=r_2e^{i\phi}$ $\Rightarrow \bar{z}=r_1e^{-i\theta}$

Given, $|z\omega|=1$ $\Rightarrow \left|r_1e^{i\theta}.r_2e^{i\phi}\right|=1$

$\Rightarrow r_1r_2=1$ ....(i)

For More Join: @IITJEE_Advanc

and $\arg(z)-\arg(\omega)=\frac{\pi}{2}$ $\Rightarrow \theta-\phi=\frac{\pi}{2}$

Then, $\bar{z}\omega = r_1 e^{-i\phi}.r_2 e^{i\phi} = r_1 r_2 e^{-i(\theta-\phi)}$

From Eqs. (i) and (ii), we get

$$\bar{z}\omega = 1.e^{-i\pi/2} = \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} \Rightarrow \bar{z}\omega = -i$$

71. Pre multiply by A

$$A^2 + A.adjA = I_2 \Rightarrow A^2 + |A|.I_2 = I_2 \Rightarrow A^2 = (1-|A|)I_2$$

$$\Rightarrow |A|^2 = (|A|-I)^2 \Rightarrow |A| = \pm(|A|-1) \Rightarrow |A| = \frac{1}{2} \Rightarrow |2A^{-1}| = 8$$

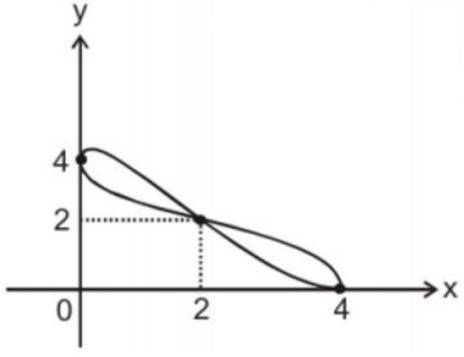
72. $$S = \frac{1}{51}\sum_{r=0}^{50}(-2)^r.{}^{51}C_{r+1}$$

$$= \frac{1}{51}\left[{}^{51}C_1(-2)^0 + {}^{51}C_2(-2)^1 + {}^{51}C_3(-2)^2 ..... + {}^{51}C_{51}(-2)^{50}\right] = \frac{1}{51}$$

73. Area $= \left|\vec{a}\times\vec{b}\right| = \left|(\vec{p}+2\vec{q})\times(2\vec{p}+\vec{q})\right|$

$$= \left|\vec{p}\times\vec{q} + 4\vec{q}\times\vec{p}\right| = \left|3\vec{p}\times\vec{q}\right| = 3\times\frac{1}{2} = \frac{3}{2}$$

74.



$$\int_0^4 f(x)-g(x)\,dx = 10 \qquad .....(1)$$

For More Join: @IITJEE_Advanc

$$\int_2^4 \left(g(x)-f(x)\right)dx = 5 \qquad \text{.....(2)}$$

From (1), $\int_0^2 \left(f(x)-g(x)\right)dx - 5 = 10$

$$\therefore \int_0^2 \left(f(x)-g(x)\right)dx = 15$$

75. Point $(3,5,7)$ satisfy the given line $L_1 = 0$

$$\therefore \frac{3-a}{l} = \frac{5-2}{3} = \frac{7-b}{4}$$

$$\Rightarrow a = 3-l, b = 3$$

Direction ratio of line joining given points is $(1,-2,-1)$ which is perpendicular to $L_1 = 0$

So, $a_1a_2 + b_1b_2 + c_1c_2 = 0$

$$\Rightarrow l - y + 4 = 0 \Rightarrow l = 2, a = 1$$

$L_1 = \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ are skew line

$$D = \left|\frac{\left(\vec{a}_2 - \vec{a}_1\right).\left(\vec{b}_1 \times \vec{b}_2\right)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| = \frac{\left(\hat{i}+2\hat{j}+2\hat{k}\right).\left(-\hat{i}+2\hat{j}-\hat{k}\right)}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

For More Join: @IITJEE_Advanc